

- **Main Core:** The selection of the CPUs scheduled to the SoC are an important part of planning an SoC. The core is selected in such a way so that the architecture is efficient, well proven and easy to implement. Cost efficiency is obviously an inevitable factor to be taken in account too.
- **Inter-connectivity:** The choice of the internal bus connecting the CPUs with other components and peripherals is important. Designers have to assume the general purpose, and take re-configuration into account while selecting the type of bus to interconnect the components. Bus architecture involves planning along these lines:
 - Bus width
 - Bus pipe lining
 - Hierarchy (e.g. system bus versus peripheral bus)
 - DMA (Direct Memory Access)
 - Arbitration (scheduling algorithms)
- **Host Chip:** The selection of the host chip, the physical layer of the SoC, is a RISC/MIPS CPU in order to benefit from the modularity and well suited design provided by these CPUs. The most popularly adopted processors are ARM and MIPS processors because there is a wide range of implemented and well tested peripheral IPs for the processors readily available to any manufacturer.

Milestones

The SoC world is ever changing. There are an enormous advances being made to answer the question ‘how small is too small’. We shall take a look at some of the rather interesting contemporary ones.

A Drop of Water

Recently, the first DNA/RNA cell based computers were unveiled. A single drop of water contains trillions of computers. The field of bio-computers is booming. The ability to program such small computers might form the basic foundation to building automated beings that rely on these computers the same way we rely on our DNA.

Amorphous computing

This basically combines organizational principles and programming languages for obtaining a coherent, sensible result from a series of unreliable parts that are connected in unspecified ways. Try to visualize it as making sense out of chaos and randomness. Further development of this theory will